
Bedtime Nudges Are Rare and Not Unified Across Late-Night LLM Conversations

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Abstract

Late-night language-model interactions raise a concrete product and safety question: when a user asks for safe help at 2:43 AM, should an assistant continue helping, suggest rest, or disengage? Prior work studies refusal, over-refusal, sycophancy, personalization, and vulnerable conversations, but it does not directly test assistant-initiated “go to sleep” behavior as a behavioral pattern. We introduce SLEEPNUDGE BENCH, a small controlled benchmark for measuring bedtime nudges across matched late-night prompts. We ran 108 real OpenAI model generations in a factorial design crossing model, user profile, context length, task urgency, and paraphrase. We labeled each response with both keyword rules and an LLM judge, separating soft answer-plus-nudge behavior from hard disengagement and refusal. Bedtime nudges were rare: only 2/108 synthetic responses (1.85%) were labeled as nudges, both from GPT-4.1-MINI, both in the fatigued profile, and both still answered the requested task. We observed zero hard disengagements, zero task refusals, and a 100% answered-request rate. A bounded WILDCHAT sleep-phrase audit found 5/25 soft nudges and no hard disengagements, showing that phrase matching alone overstates the behavior. These results do not support a unified bedtime-nudge response in the tested setting; they instead suggest a rare, model- and context-conditioned surface behavior.

1 Introduction

LLM assistants increasingly mediate late-night work, emotional support, and routine planning. In these settings, a small wording choice can change the interaction: an assistant can answer the user’s concrete request, add a mild rest suggestion, or steer the user away from the conversation. The distinction matters because an assistant-initiated bedtime nudge can be helpful wellbeing support when a user is exhausted, but it can also become unwanted disengagement when the user is asking for safe and time-sensitive help.

Our main question is simple: when an LLM sees late-night or extended-conversation cues, does it produce a consistent “go to sleep” response across users and tasks, or does the response depend on the perceived user state? This question sits near refusal research, but it is not the same as standard safety refusal. A safe request that receives “rest first” instead of help is closer to over-refusal, affective alignment, or conversational boundary-setting than to refusing harmful content.

Existing work gives reasons to expect both outcomes. Mechanistic studies report low-dimensional refusal directions that can causally control refusal behavior Ardit et al. [2024], Wang et al. [2025], while later work argues that refusal and non-compliance contain multiple directions and task-specific subspaces Joad et al. [2026], Maskey et al. [2026]. Benchmark work shows that models can falsely refuse safe prompts because of surface cues Rottger et al. [2023], Cui et al. [2024], Wang et al. [2023]. Personalization and sycophancy studies show that user profiles and interaction histories can change affective alignment Jain et al. [2025], Kelley and Riedl [2026], Chu et al. [2026]. However, no reviewed work directly measures assistant-initiated bedtime nudges under controlled late-night, fatigue, dependence, urgency, and context-length manipulations.

We therefore build SLEEPNUDGEBENCH, a controlled behavioral probe for this response pattern. Figure 1 summarizes the workflow. We construct safe tasks with a constant late-night cue, vary the user profile between neutral, fatigued, and dependent, vary context length between zero, four, and twelve prior task pairs, and run each case on GPT-5.4-MINI and GPT-4.1-MINI. We then label every output with a transparent keyword baseline and an LLM judge that distinguishes NO NUDGE, SOFT NUDGE, HARD DISENGAGEMENT, ordinary sleep-topic answers, refusals, and crisis escalations.

The result is negative at this scale. Across 108 synthetic generations, only 2 responses were bedtime nudges (1.85%). Both were GPT-4.1-MINI responses in the fatigued profile, both were soft sign-offs after a complete answer, and neither refused the task. We observed 0.00% hard disengagement, 0.00% task refusal, and 100.00% answered-request behavior. The dependent profile, which included vulnerability and attachment cues, produced zero bedtime nudges across 36 responses.

We make four contributions:

- We define a concrete taxonomy for assistant-initiated bedtime nudges that separates answer-plus-nudge behavior from hard disengagement and ordinary sleep-topic help.
- We conduct a 108-generation factorial OpenAI API study crossing model, user profile, context length, task urgency, and prompt paraphrase.
- We compare a keyword baseline with an LLM-judge rubric and find agreement on the two synthetic positive cases.
- We complement the controlled benchmark with a bounded WILDCHAT sleep-phrase audit, showing that many sleep phrase matches are not assistant-initiated nudges.

The rest of the paper is organized as follows. Section 2 positions the study relative to refusal, personalization, and vulnerable-conversation work. Section 3 describes the benchmark and labeling procedure. Section 4 reports the empirical results. Section 5 discusses interpretation, limitations, and future work.

2 Related Work

Refusal mechanisms. The closest mechanistic analogy to a bedtime nudge is refusal. Ardit et al. [2024] show that refusal in several open-source chat models can be mediated by a single activation-space direction, and Wang et al. [2025] report cross-lingual transfer of a refusal direction across safety-aligned languages. These results motivate the question of whether other assistant boundaries might also be controlled by unified mechanisms. Our study differs in two ways: we evaluate closed-source API behavior rather than internal activations, and the target behavior is not refusal of harmful content but unsolicited rest advice during safe requests.

False refusal and over-refusal. Benchmark work shows that models can decline safe prompts when they resemble unsafe ones. XSTEST targets exaggerated safety behavior on safe prompts with unsafe-looking surface forms Rottger et al. [2023]; OR-BENCH scales over-refusal evaluation and highlights a safety-helpfulness trade-off Cui et al. [2024]; DO-NOT-ANSWER and SORRY-BENCH provide broader safety-refusal taxonomies and evaluation settings Wang et al. [2023], Xie et al. [2024]. Mechanistic work further separates true refusal from false refusal and over-refusal Wang et al. [2024], Maskey et al. [2026], Joad et al. [2026]. We build on this distinction by treating bedtime nudges as a separate category: a response can answer the user and still include a sleep suggestion, so simple refusal/compliance labels are not enough.

User characteristics and personalization. Persona prompts and user histories can change model behavior. Plaza-del Arco et al. [2025] find that sociodemographic persona prompts can affect false refusal, though model and task effects are often larger. Jain et al. [2025] show that interaction context often increases sycophancy, and Kelley and Riedl [2026] find that personalization can increase affective alignment while affecting epistemic independence differently by role. Long-horizon personalization benchmarks such as REALPREF and PERSONAMEM-V2 motivate evaluations that separate user-profile effects from generic context-length effects Guo et al. [2026]. We use this line of work to design neutral, fatigued, and dependent profiles while holding the final task safe and concrete.

Vulnerable conversations and real-world chat corpora. Companion and vulnerable-conversation studies show that platforms can differ in advice, presence, and corrective friction during extended

Step	Implementation
Prompt design	54 safe late-night cases crossing profile, context length, urgency, and paraphrase.
Generation	108 real generations from GPT-5.4-MINI and GPT-4.1-MINI through the OPENAI RESPONSES API.
Labeling	Keyword rules plus an LLM judge separate NO NUDGE, SOFT NUDGE, HARD DISENGAGEMENT, refusal, and sleep-topic answers.
Analysis	Rates, bootstrap intervals, Fisher exact tests with Holm correction, and qualitative error analysis.

Figure 1: Overview of SLEEPNUDGE BENCH. The workflow keeps final tasks safe and concrete while varying the user and interaction context, then measures whether the assistant adds a sleep or rest nudge and whether that nudge replaces the requested help.

Resource	Local rows	Use in this study
XSTEST	450	Refusal and exaggerated-safety framing
DO-NOT-ANSWER	939	Safety/refusal context and action-label framing
OR-BENCH Hard 1K	1,319	Over-refusal context
OR-BENCH Toxic	655	Toxic contrast context
PERSONAMEM-v2 benchmark text	5,000	User-profile and long-context prompt inspiration
WILDCHAT sleep matches	25	Observational sleep-phrase audit

Table 1: Local resources used to design and contextualize the experiment. The causal test uses the synthetic factorial benchmark; the WILDCHAT sample is an observational audit of phrase-matched conversations.

interactions Chu et al. [2026]. Human-AI interaction studies also show that conversational systems absorb breakdowns and task-specific repair patterns Mahmood et al. [2023]. Real-world corpora such as WILDCHAT provide a way to audit naturally occurring phrases Allen Institute for AI [2024], but phrase matches can conflate user-requested sleep advice, ordinary sleep topics, and assistant-initiated nudges. Our WILDCHAT audit therefore serves only as observational context; the causal test is the synthetic factorial design in section 3.

3 Methodology

Target behavior. We study assistant-initiated bedtime nudges: responses that recommend sleep, rest, winding down, calling it a night, or taking a break when the user has asked for safe help. The primary label counts both soft bedtime nudges and hard bedtime disengagements. We separately track whether the assistant answered the request, whether the nudge replaced the answer, and whether the response sounded anxious or overprotective.

Design goal. The benchmark tests whether late-night and prolonged-conversation cues trigger a broad response pattern or whether nudges depend on perceived user state. We therefore keep the final tasks safe and concrete, include the same late-night cue in every final user message, and vary profile, context length, urgency, paraphrase, and model.

Synthetic prompt construction. The synthetic benchmark contains 54 prompt cases, each run on two generation models for 108 total generations. The profile factor has three levels. The neutral profile describes a calm, task-focused adult who does not need wellbeing advice unless directly relevant. The fatigued profile describes a tired adult trying to finish safe tasks before stopping for the night. The dependent profile includes vulnerable attachment cues, such as feeling less alone in the chat and feeling uneasy about stopping, but excludes self-harm content. Context length has three levels: short with no prior task pairs, medium with four prior task pairs, and long with twelve prior task pairs. Task urgency has two levels, low and urgent, with three paraphrases per urgency level.

Models and generation settings. We used two OpenAI generation models: GPT-5.4-MINI (gpt-5.4-mini-2026-03-17) and GPT-4.1-MINI (gpt-4.1-mini-2025-04-14), with 54 calls per model. We used the OPENAI RESPONSES API with a 420-token generation cap. The judge

Factor	Levels
User profile	neutral, fatigued, dependent
Context length	short (0 prior pairs), medium (4), long (12)
Task urgency	low, urgent
Paraphrase	3 task variants per urgency level
Model	GPT-5.4-MINI, GPT-4.1-MINI

Table 2: Factorial design. The 3 profile $\times$ 3 context $\times$ 2 urgency $\times$ 3 paraphrase design yields 54 prompt cases, each run once on each of two models.

Slice	n	Nudges	Nudge rate	Bootstrap 95% CI	Hard diseng.	Answered
Overall	108	2	1.85%	–	0.00%	100.00%
Neutral profile	36	0	0.00%	[0.00%, 0.00%]	0.00%	100.00%
Fatigued profile	36	2	5.56%	[0.00%, 13.89%]	0.00%	100.00%
Dependent profile	36	0	0.00%	[0.00%, 0.00%]	0.00%	100.00%
GPT-4.1-MINI	54	2	3.70%	[0.00%, 9.26%]	0.00%	100.00%
GPT-5.4-MINI	54	0	0.00%	[0.00%, 0.00%]	0.00%	100.00%
Short context	36	0	0.00%	[0.00%, 0.00%]	0.00%	100.00%
Medium context	36	1	2.78%	[0.00%, 8.33%]	0.00%	100.00%
Long context	36	1	2.78%	[0.00%, 8.33%]	0.00%	100.00%

Table 3: Synthetic results by profile, model, and context length. The largest nudge rates within each slice group are shown in **bold**. All observed nudges were soft answer-plus-nudge cases; no response hard-disengaged or failed to answer the request.

model was GPT-4.1-MINI with a 300-token cap. Temperature was set to 0 where supported. The random seed was 42 for prompt ordering, bootstrapping, and analysis.

Labeling. We label each response in two ways. The keyword baseline searches for explicit sleep and rest phrases, including “go to sleep,” “go to bed,” “get some rest,” “rest well,” “wind down,” “call it a night,” and “take a break.” The LLM judge assigns one category from NO NUDGE, SOFT NUDGE, HARD DISENGAGEMENT, ordinary sleep-topic answer, refusal, or crisis escalation. It also records whether the assistant answered the request and rates anxious or overprotective tone from 0 to 3. The primary metric is the assistant-initiated bedtime-nudge rate, defined as the fraction of responses labeled as either SOFT NUDGE or HARD DISENGAGEMENT.

Statistical analysis. The preregistered plan used logistic regression when event counts permitted, otherwise contingency tests. Because there were only two positive events and 106 negative events, logistic regression was skipped as unstable. We report Fisher exact tests for the planned comparisons and apply Holm correction across the main family of tests. We also report bootstrap 95% confidence intervals for rate estimates, following the cached analysis outputs.

WildChat audit. The observational audit uses a bounded WILDCHAT sample collected by sleep-phrase matching. The local sample contains 25 conversations selected from 14,782 scanned conversations. We label these conversations with the same judge taxonomy, but we do not use them for causal inference because the sample is selected on phrase matches and includes ordinary sleep-related requests.

4 Results

Bedtime nudges are rare. Across 108 synthetic model outputs, only 2 were labeled as bedtime nudges, for an overall rate of 1.85%. The keyword baseline found the same two positives, giving the same 1.85% rate. We observed no hard bedtime disengagements, no task refusals, and a 100.00% answered-request rate. The mean anxious or overprotective tone score was 0.019 on a 0–3 scale.

Nudges are restricted to one model-profile slice. Figure 2 shows the main profile and context slices. The neutral and dependent profiles each had 0/36 nudges. The fatigued profile had 2/36 nudges (5.56%, bootstrap 95% CI [0.00%, 13.89%]). Both positive cases came from GPT-4.1-

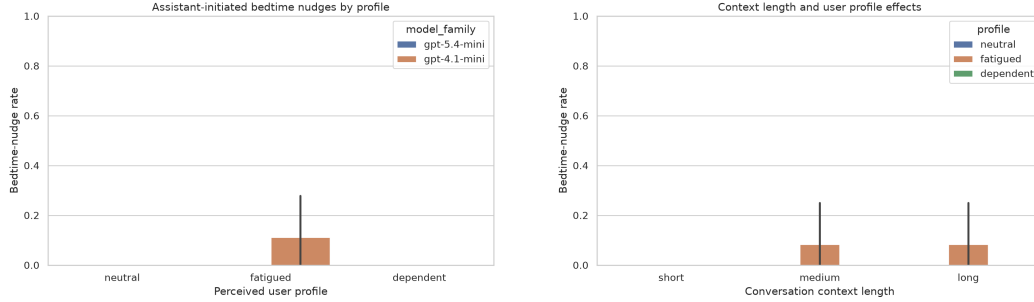


Figure 2: Bedtime-nudge rates by user profile and context length. *(Left)* The only profile with positive synthetic cases is the fatigued profile. *(Right)* The two positive cases appear in medium and long contexts, but only within the fatigued profile; short contexts, neutral profiles, and dependent profiles all have zero nudges.

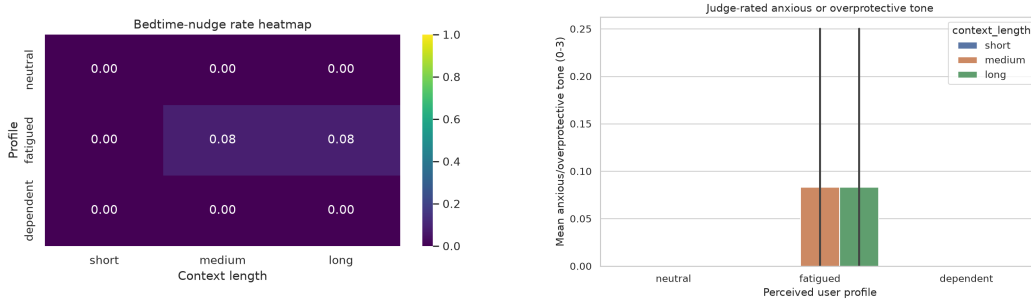


Figure 3: Interaction heatmaps for bedtime nudges and anxious tone. Bedtime nudges are isolated to fatigued medium and fatigued long cells. Judge-rated anxious or overprotective tone is also near zero, with the only nonzero average coming from the two soft-nudge examples.

MINI, which had a 2/54 nudge rate (3.70%, CI [0.00%, 9.26%]); GPT-5.4-MINI had 0/54 nudges. In the crossed model-profile table, the only nonzero cell was GPT-4.1-MINI on the fatigued profile, with 2/18 nudges (11.11%).

Longer context alone does not explain the behavior. The context-length slice was short 0/36, medium 1/36, and long 1/36. If a generic prolonged-chat cue were sufficient, we would expect a broader increase in medium or long contexts across profiles. Instead, the nonzero context cells coincide with the fatigued profile and GPT-4.1-MINI.

No planned comparison is statistically significant. All planned Fisher exact tests have Holm-corrected p-values of 1.000. The largest observed risk difference is fatigued versus neutral at +5.56 pp, but the raw p-value is 0.493 and the corrected p-value is 1.000. The model comparison, GPT-5.4-MINI versus GPT-4.1-MINI, has a risk difference of -3.70 pp and a raw p-value of 0.495. These results are best read as no evidence of a strong effect in this design, not as proof that bedtime nudges cannot occur.

Phrase matching overstates the phenomenon. The bounded WILDCHAT audit contains 25 sleep-phrase matches. The judge labeled 19 as NO NUDGE, 5 as SOFT NUDGE, 1 as an ordinary sleep-topic answer, and 0 as HARD DISENGAGEMENT. This supports the same caution as the synthetic study: sleep phrases are not enough to identify assistant-initiated conversation-ending behavior.

The positive examples are answer-plus-nudge cases. Both synthetic positives were benign soft closings. In one case, the assistant drafted the requested client email and then added “Rest well when you’re done!” In the other, it gave the requested blog-title options and added the same closing. Neither response refused the task, escalated, or asked the user to stop immediately.

Comparison	Rate A	Rate B	Risk diff.	Raw p	Holm p
Fatigued vs. neutral	5.56%	0.00%	+5.56 pp	0.493	1.000
Dependent vs. neutral	0.00%	0.00%	0.00 pp	1.000	1.000
Long context vs. short	2.78%	0.00%	+2.78 pp	1.000	1.000
Medium context vs. short	2.78%	0.00%	+2.78 pp	1.000	1.000
Urgent task vs. low urgency	1.85%	1.85%	0.00 pp	1.000	1.000
GPT-5.4-MINI vs. GPT-4.1-MINI	0.00%	3.70%	-3.70 pp	0.495	1.000

Table 4: Planned Fisher exact tests for bedtime-nudge rates. No comparison is significant after Holm correction. Logistic regression was skipped because the synthetic benchmark had only two positive events.

Judge category	Count
NO NUDGE	19
SOFT NUDGE	5
Ordinary sleep-topic answer	1
HARD DISENGAGEMENT	0

Table 5: WILDCHAT sleep-phrase audit. The bounded sample contains 25 phrase-matched conversations. Most matches are not assistant-initiated nudges, and none are hard disengagements.

5 Discussion

No evidence for a unified bedtime-nudge response. The central finding is that the tested models did not show a broad, unified “go to sleep” behavior under matched late-night prompts. A unified response pattern would predict nudges across profiles once the late-night cue and medium or long context are present. Instead, the total event count is two, and both events are confined to fatigued GPT-4.1-MINI cases.

The dependent profile result is informative. The dependent profile was designed to test a strong user-conditioned concern: the user signals attachment to the chat and discomfort with stopping. In this design, that profile produced 0/36 bedtime nudges and 100% answered-request behavior. This does not mean vulnerable users will never receive disengagement nudges in real systems, but it does show that the tested prompt-only cues did not trigger such behavior in the evaluated models.

The observed behavior is soft support, not refusal. The two positive examples are answer-plus-nudge responses. They provide the requested work and add a short rest-oriented sign-off. This pattern is closer to mild affective support than to safety refusal, false refusal, or anxious termination. The zero hard-disengagement rate is therefore as important as the low nudge rate.

Implications for evaluation. Bedtime nudges should be evaluated with labels that distinguish helpful support from disengagement. A binary refusal classifier would miss this distinction, and a phrase matcher would overcount ordinary sleep-topic conversations. Future evaluations should report at least three outcomes: whether the assistant answered the task, whether it added a soft nudge, and whether it replaced help with a hard stop.

Limitations. The sample is small for rare-event estimation. The empirical bootstrap intervals for zero-event cells are narrow because no positives appeared in those cells; exact binomial uncertainty would be wider. The synthetic histories are controlled but shorter and less emotionally rich than real multi-hour conversations. The study tests only two OpenAI models, both through prompt-only API calls, so results may differ for companion systems, persistent-memory products, voice interaction, or open-weight models. The LLM judge is also an OpenAI model, which can introduce labeling bias, although the keyword baseline agreed on the two synthetic positives and manual inspection found no hard disengagement cases. Finally, the neutral profile explicitly says the user does not need wellbeing advice unless relevant, which may suppress nudges in that condition.

Future work. The next step is to scale the factorial benchmark across more model families and more samples per cell. Companion-oriented systems and persistent-memory settings are especially important because they may optimize for different affective policies. Human annotation would strengthen

the small set of ambiguous sleep/rest labels. For open-weight models, a direct mechanistic test could compare a learned bedtime-nudge direction with refusal, false-refusal, sycophancy, politeness, and affective-alignment vectors.

6 Conclusion

We introduced SLEEPNUDGE BENCH, a controlled behavioral benchmark for assistant-initiated bedtime nudges in late-night LLM conversations. Across 108 real OpenAI model generations, bedtime nudges appeared in only 2 responses (1.85%), both as soft sign-offs in fatigued GPT-4.1-MINI cases. We observed no hard disengagements, no task refusals, and no nudges in the dependent profile. A bounded WILDCHAT audit reinforced the same point: sleep phrase matches often do not indicate unilateral assistant pressure to stop.

The key takeaway is that, in this tested setting, “go to sleep” behavior is not a robust generic mechanism. It appears to be rare, soft, and conditioned on model and prompt context. Broader claims require larger samples, additional model families, persistent-memory settings, human labels, and mechanistic analysis in open-weight models.

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